

Computer Networks Lab

Fall 2025

Week 10

Dynamic routing using RIP

What is Dynamic Routing?

Dynamic routing is a network routing method in which routers automatically adjust their paths for data transmission based on real-time changes in the network topology. Unlike static routing where routes are manually configured, dynamic routing protocols enable routers to:

- Communicate with each other
- Share network information automatically
- Determine the best paths for data packets
- Adapt to network changes without manual intervention

Advantages of Dynamic Routing:

1. **Automatic Route Updates:** Routes are learned and updated automatically
2. **Scalability:** Suitable for large networks with many routers
3. **Fault Tolerance:** Automatically finds alternate paths if a link fails
4. **Reduced Administrative Overhead:** No need to manually configure every route
5. **Load Balancing:** Can distribute traffic across multiple paths

Disadvantages of Dynamic Routing

1. **Resource Consumption:** Uses CPU, memory, and bandwidth for routing updates
2. **Complexity:** More complex to configure and troubleshoot than static routing
3. **Convergence Time:** Takes time for all routers to agree on network topology
4. **Security Concerns:** Routing updates can be intercepted or manipulated

Types of Dynamic Routing Protocols

Dynamic routing protocols are classified into three main categories:

1. Distance Vector Protocols

- Routes are determined by the number of hops (distance) between source and destination
- Each router shares its routing table with directly connected neighbors
- Examples: **RIP (Routing Information Protocol)**, IGRP (Interior Gateway Routing Protocol)
- Uses periodic updates (every 30 seconds for RIP)

2. Link-State Protocols

- Routers share information about the state of their links (bandwidth, cost, delay)
- Each router builds a complete topology map of the network
- Examples: **OSPF (Open Shortest Path First)**, IS-IS (Intermediate System to Intermediate System)
- Uses triggered updates (only when topology changes)

3. Hybrid Protocols

- Combines features of both distance vector and link-state protocols
- Example: **EIGRP (Enhanced Interior Gateway Routing Protocol)**

RIP (Routing Information Protocol) Overview

RIP is one of the oldest distance vector routing protocols and is widely used for teaching purposes due to its simplicity.

Key Characteristics of RIP:

Feature	Details
Metric	Hop count (maximum 15 hops, 16 = unreachable)
Algorithm	Bellman-Ford algorithm
Update Timer	30 seconds (periodic updates)
Administrative Distance	120
Protocol	Uses UDP port 520
Versions	RIPv1 (classful), RIPv2 (classless - supports VLSM)

RIP Timers:

1. **Update Timer (30 seconds):** How often routing updates are sent
2. **Invalid Timer (180 seconds):** Route is marked invalid if no update received
3. **Flush Timer (240 seconds):** Route is removed from routing table

How RIP Works:

1. Router sends its entire routing table to directly connected neighbors
2. Neighboring routers receive the update and add routes to their tables
3. Each router increments the hop count by 1 for received routes
4. Process repeats every 30 seconds
5. Routers keep the route with the lowest hop count

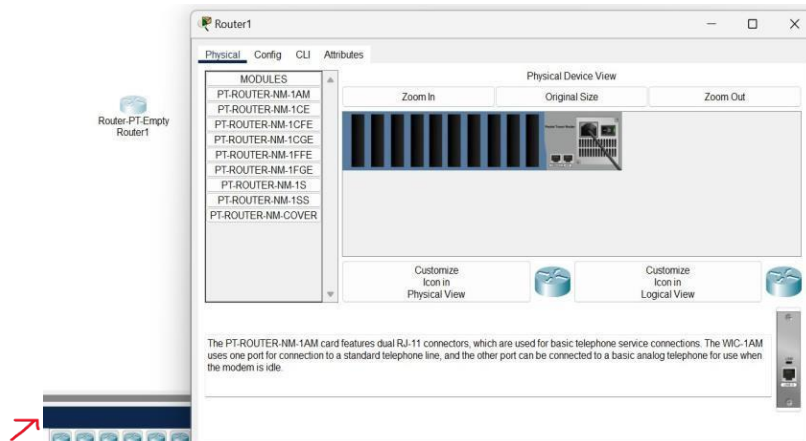
Walk through:

Setting up the router:

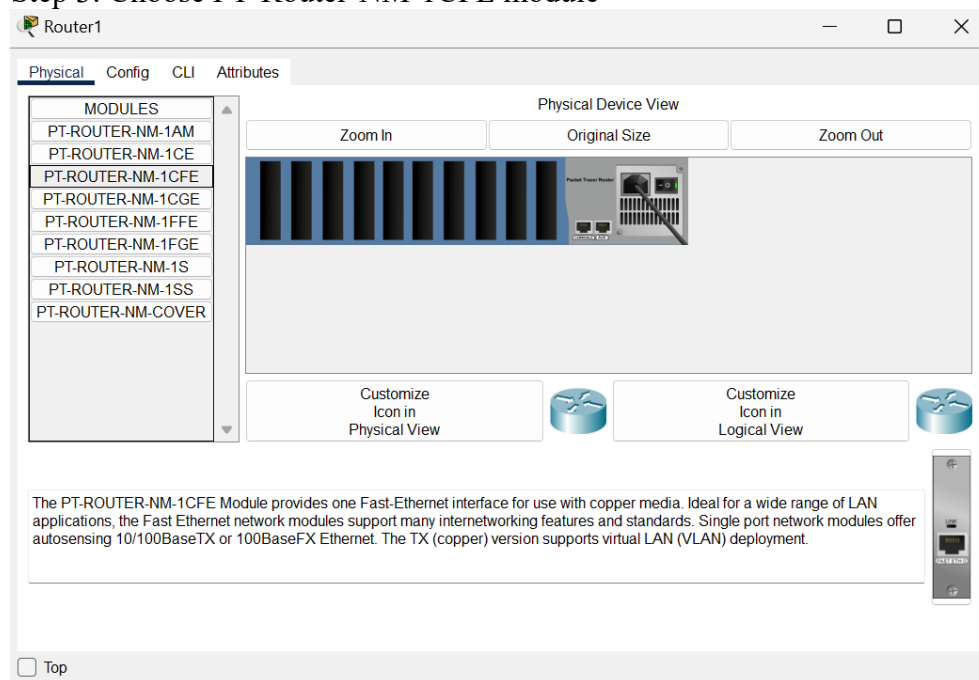
Step 1: Select router PT-Empty from the network devices, and place it to your work space.



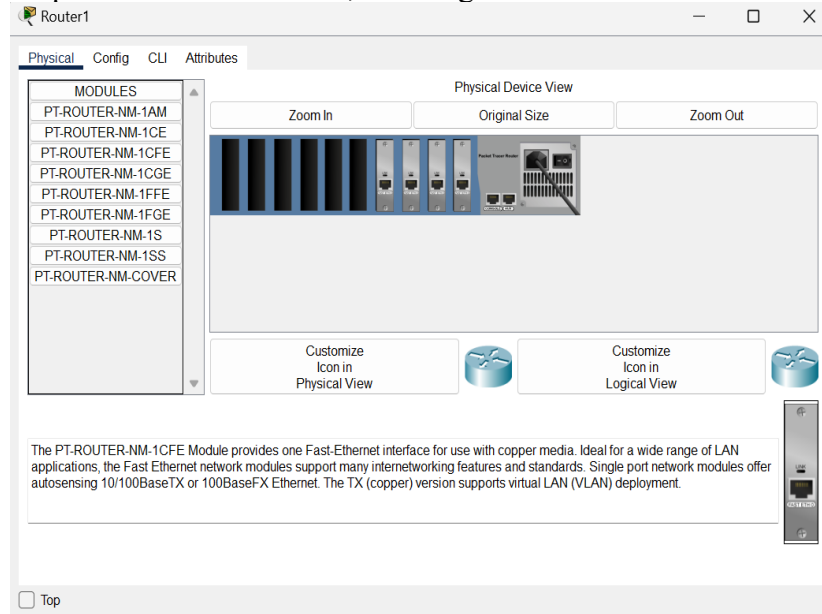
Step 2: Click on the router to get its physical interface



Step 3: Choose PT-Router-NM-1CFE module



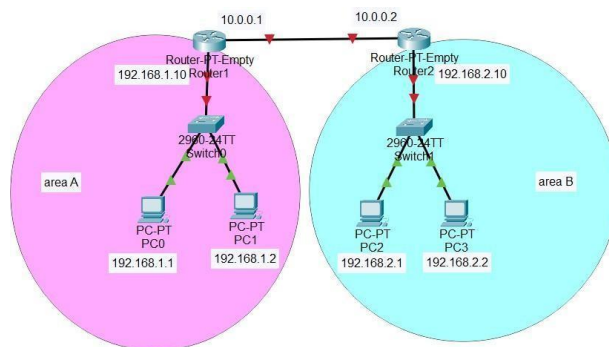
Step 4: Turn off the router, and drag the module and fill in the empty slots



Step 5: Turn on the router



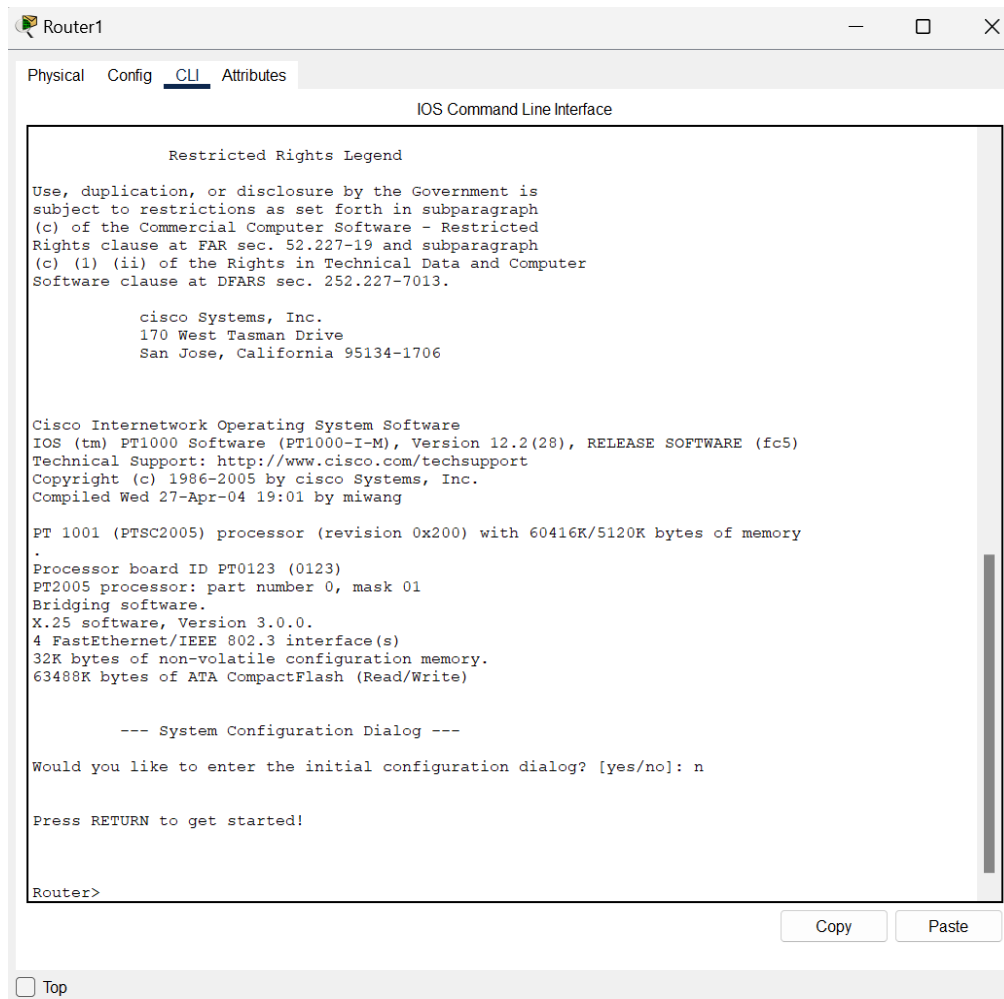
Step 6: Construct the given topology



Configure the router:

Step 1: In area A, the router and switch are attached to port fastethernet0/0 of the router. Go to the router attached in the area A, go to its CLI.

Step 2: type 'no' to the question: Would you like to enter the initial configuration dialog? [yes/no]



Step 3: type 'en'

```
Router>en
Router#
```

Step 4: type 'config t'

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
```

Step 5: type 'interface fa0/0' for access fastether0/0 interface

```
Router(config)#
Router(config)#interface fa0/0
Router(config-if)#
```

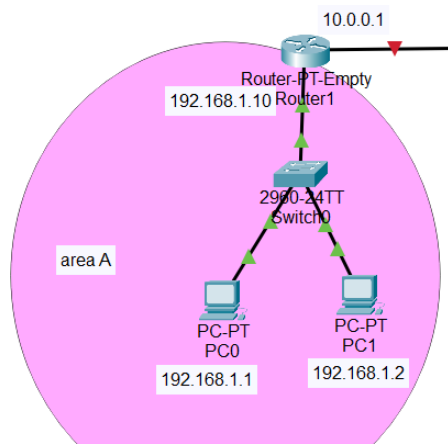
Step 6: time to enter the ip address to the port. Write 'ip address' then the ip address and the subnetmask

```
Router(config)#  
Router(config)#interface fa0/0  
Router(config-if)#ip address 192.168.1.10 255.255.255.0  
Router(config-if)#
```

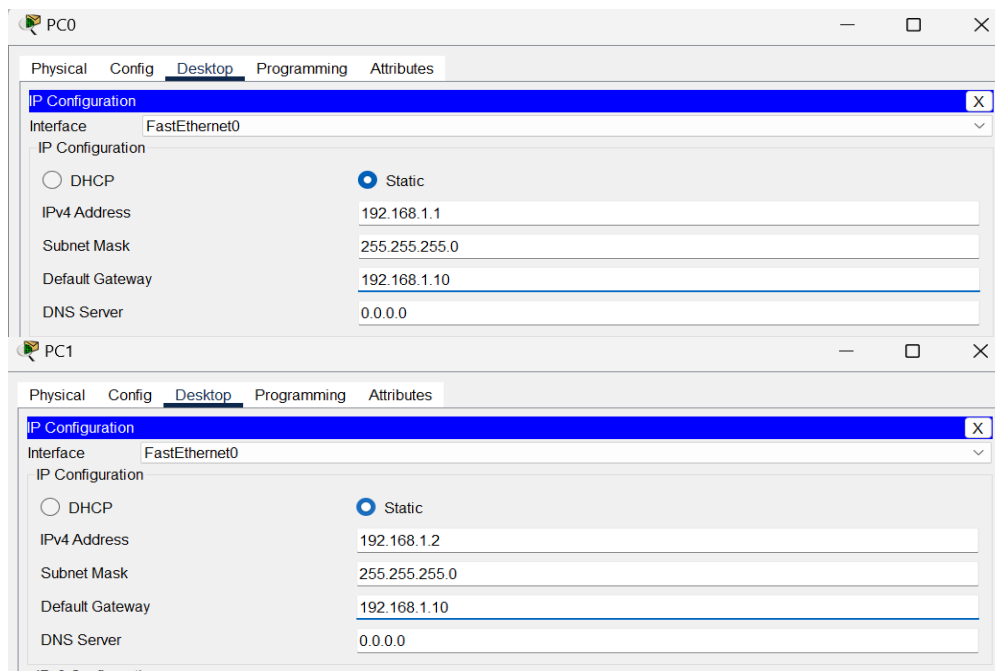
Step 7: type 'no shut' to turn on the port

```
Router(config-if)#  
Router(config-if)#no shut  
  
Router(config-if)#  
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up  
Router(config-if)#
```

Step 8: The port will be up after the steps



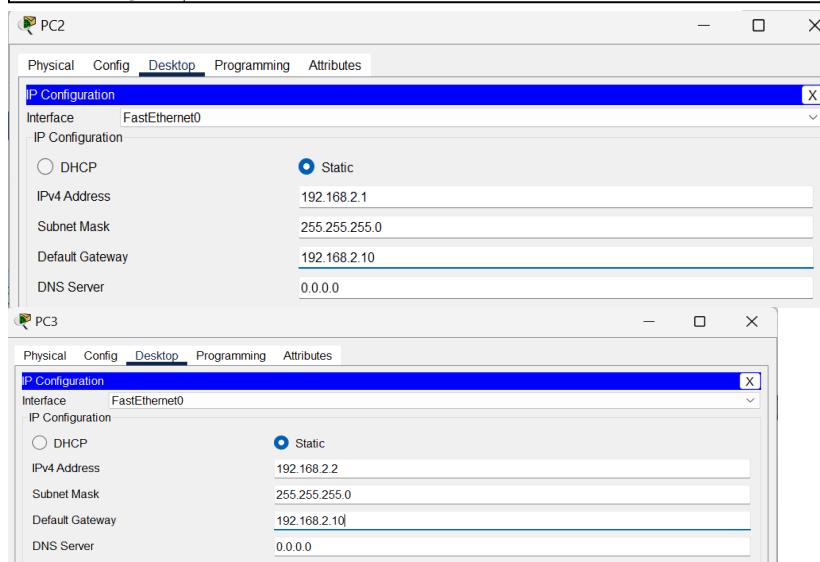
Step 9: Assign ip address, subnet mask, and default gateway to the PCs in area A. Default gateway will be the ip address assigned to the router's interface connecting the switch in area A



Step 10: repeat all the steps for area B configuration

```
Router>
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.2.10 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#
```



Step 11: to configure the network in between the two routers, go to router1's CLI. Configure the port connecting the two routers together.

Step 12: Assign ip address to the port at router1

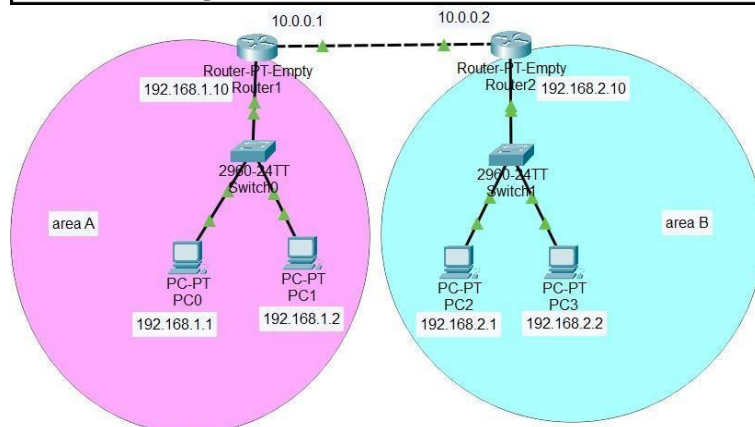
```
Router>
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa1/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
Router(config-if)#
```

Step 13: repeat the same configuration at the other router as well, with different ip address

```
Router>
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa1/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
Router(config-if)#
```



Configure routing

Step 1: in router1, go to its CLI and write the command 'router rip'

Step 2: enter all the networks directly connected to the specific router one by one as mentioned below

```
Router(config)#
Router(config)#router rip
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#
```

Step 3: in router2, repeat the above steps


```

Router(config) #
Router(config)#router rip
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router) #

```

Step 4: once done, pick the inspect option and click the router1 to get details of the routing table. If the information of all the networks appears in the list then you are good to go, else check for errors. Check for router2 as well.

Type	Network	Port	Next Hop IP	Metric
C	10.0.0.0/8	FastEthernet1/0	---	0/0
C	192.168.1.0/24	FastEthernet0/0	---	0/0
R	192.168.2.0/24	FastEthernet1/0	10.0.0.2	120/1

Type	Network	Port	Next Hop IP	Metric
C	10.0.0.0/8	FastEthernet1/0	---	0/0
R	192.168.1.0/24	FastEthernet1/0	10.0.0.1	120/1
C	192.168.2.0/24	FastEthernet0/0	---	0/0

Step 5: send pdu to the other network to check the working of the topology

Simulation Panel

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	Router1	ICMP
	0.003	Router1	Router2	ICMP
	0.004	Router2	Switch1	ICMP
	0.005	Switch1	PC2	ICMP
	0.006	PC2	Switch1	ICMP
	0.007	Switch1	Router2	ICMP
	0.008	Router2	Router1	ICMP
	0.009	Router1	Switch0	ICMP
	0.010	Switch0	PC0	ICMP

Reset Simulation ☒ Constant Delay Captured to 0.010

Play Controls: [Previous] [Play] [Next] [Stop]

Event List Filters - Visible Events: ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIP, RIPng, RTI, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USI, VTP

Edit Filters Show All/None

Scenario 0: [New] [Delete] [Fire] Last Status: Successful Source: PC0 Destination: PC2 Type: ICMP Color: [Purple] Time(sec): 0.000 Periodic: N Num: 0 Edit: (edit) Delete: (delete)